Bishwa Ranjan Si

Experimental Mechanobiology | *Hydrogel and Polymer Systems* | *Quantitative Fluorescence Microscopy* | *Optogenetics*
Baltimore, MD, USA · bsi1@jhu.edu · LinkedIn · Google Scholar

PROFESSIONAL SUMMARY

Experimentalist with 7+ years of research in quantitative imaging, hydrogels, polymer systems and cell mechanobiology. Expertise in mammalian cell culture, molecular biology, tunable hydrogels, immunostaining, confocal/live-cell imaging, optogenetics, zebrafish xenografts, and image analysis. 10 publications (5 first/co-first author), including *Science Advances, Nature Cell Biology* and *Cancer Research*.

EDUCATION

Ph.D., Chemical and Biomolecular Engineering *Expected Dec 2026*
Johns Hopkins University, Baltimore, MD; Advisor: **Prof. Konstantinos Konstantopoulos**

B.S. and M.S., Chemical Engineering *2014 – 2019*
Indian Institute of Technology Kanpur, India; Advisor: **Prof. Rahul Mangal**

TECHNICAL SKILLS

Hydrogels and Polymers	polyacrylamide hydrogel formulation (tunable stiffness), collagen gel embedding, gel rheology, polymer–nanoparticle composites, thin-film coating, microfluidic fabrication
Sample Preparation	mammalian cell culture, 3D tumor spheroids, immunofluorescence, zebrafish xenograft microinjection and imaging
Microscopy	confocal and live-cell imaging, fluorescence lifetime imaging (FLIM), traction force microscopy, optogenetics, SEM, optical microscopy, acquisition optimization, image-quality assessment
Image and Data Analysis	Python, MATLAB, Fiji/ImageJ, IMARIS, GraphPad Prism; segmentation, cell tracking, line-scan, polarization, and reproducibility analysis
Molecular Biology	lentiviral transduction, plasmid design, RNAi, CRISPR, RT-qPCR, Western blot, flow cytometry
Metabolism Assays	Seahorse analysis, mitochondrial isolation/transplantation, metabolic biosensors

RESEARCH EXPERIENCE

Graduate Research Assistant, Johns Hopkins University, Baltimore, MD *2021 – Present*

- Led studies of 4 microenvironmental cues (viscosity, confinement, stiffness, electric fields) in cancer cell migration; showed elevated viscosity enhances 3D invasion via KCC4-dependent signaling.
- Formulated tunable polyacrylamide hydrogels matching healthy and tumor-associated breast tissue stiffness; characterized gels by rheology to link matrix mechanics to cell behavior.
- Optimized fixation, permeabilization, immunolabeling, and confocal acquisition for 2D cultures and 3D samples, preserving morphology and signal specificity for quantitative measurement.
- Mapped 3 ion channels (NHE1, KCC4, SWELL1) and cytoskeletal regulators by immunofluorescence and live-cell imaging, using ROI, line-scan, and polarization analysis.
- Engineered optogenetic constructs for 3 modes of targeted protein control (activation, recruitment, degradation), tuning expression and illumination for live-cell confocal quantification.
- Developed 5 quantitative QC metrics (protein localization, cell tracking, migration velocity, direction reversals, 3D dissemination) to benchmark reproducibility across experiments.
- Standardized workflows across 4 model systems (2D culture, microfluidic migration assays, 3D collagen-embedded spheroids, zebrafish xenografts), with tumor cell injection at 3 different zebrafish sites.
- Designed a microfluidic electrotaxis device to probe electric-field effects on confined cancer cell migration, contractility, and calcium signaling (3 conference presentations: manuscript in preparation).
- Generated decision-driving data and contributed to grant writing for one funded NIH R01 and three additional NIH R01 applications, while supporting therapeutic target discovery.

Project Associate, Indian Institute of Technology Kanpur, India *Jun 2020 – May 2021*

- Investigated active liquid-crystal droplets and Janus colloids in complex fluids using rheology and optical microscopy; first-authored an Elsevier book chapter.

Post-graduate Engineer, Sterlite Technologies, Pune, India *Jul 2019 – Apr 2020*

- Completed a three-month training program in optical fiber manufacturing and cabling plants, then identified key process improvements to reduce downtime and enhance operational efficiency of the fiber drawing plant.

M.S. Research Assistant, Indian Institute of Technology Kanpur, India *2017 – 2019*

- Investigated active Janus colloids and active liquid-crystal droplets using rheology, optical microscopy, SEM & thin-film coating to study the hydrodynamic and interfacial effects on their motion. (2 papers: *Langmuir*, *Physics of Fluids*).

PUBLICATIONS

*156 citations, h-index 4, i10-index 4 (Google Scholar, Oct 2026). *Equal contribution.*

First or co-first author

- **Si BR***, Choudhury D*, et al. "Extracellular Fluid Viscosity Regulates Cancer Cell Invasion Through KCC4-Dependent Signaling." *Science Advances* (accepted, 2026).
- **Si BR***, Choudhury D*, Nath S*, Konstantopoulos K. "Galvanin: a molecular compass for bioelectric navigation." *Cell Research* (Research Highlight, 2026). DOI
- **Si BR**, Yuan Q, Nath S, et al. "Engineered In Vitro Platforms for Mechanochemical Control of Cell Migration." *Cold Spring Harbor Perspectives in Biology* 17, a041795 (2025). DOI
- **Si BR**, Mangal R. "Morphology of Nanoparticle-Based Polymer Composites." In *Nanoparticle-Based Polymer Composites*, pp. 151–179, Elsevier (2022). DOI
- **Si BR**, Patel P, Mangal R. "Self-Propelled Janus Colloids in Shear Flow." *Langmuir* 36, 11888–11898 (2020). DOI
- Sen A*, **Si BR***, Choudhury D*, et al. "Viscous Memory Regulates Cancer Cell Invasion *In Vitro* and *In Vivo*." (submitted)
- **Si BR**, et al. "Electric Field-Guided Migration of Confined Cancer Cells: Roles of Cellular Contractility and Calcium Signaling." (in preparation).

Co-author

- Amitrano A*, …, **Si BR**, et al. "Mechanochemical Cues Control the Coupling of Metabolic and Migratory Patterns in Cancer." *Nature Cell Biology* (accepted, 2026). Preprint
- Amitrano A, … **Si BR**, et al. Rethinking Tumor Acidosis: Driver, Brake, or Adaptive Challenge? *American Journal of Physiology–Cell Physiology* (invited).
- Stoletov K, …, **Si BR**, et al. "Adrenomedullin 2 Drives Intratumoral Cancer Cell Invasion, Intravasation, and Metastasis." *Oncogene* 45, 4916–4926(2026). DOI
- Ge Z, Ni Q, Yan Y, …, **Si BR**, et al. "Anionic Lipid Trafficking and Cell Mechanics Regulate Membrane Electrical Potential in Non-Excitable Tissue Cells." *bioRxiv* (2025); under review at *Nature Materials*. DOI
- Mistriotis P, Wisniewski EO, **Si BR**, Kalab P, Konstantopoulos K. "Coordinated in confined migration: crosstalk between the nucleus and ion channel-mediated mechanosensation." *Trends in Cell Biology* 34, 809–825 (2024). DOI
- Lee G, Wong C, Cho A, …, **Si BR**, et al. "E-Cadherin Induces Serine Synthesis to Support Progression and Metastasis of Breast Cancer." *Cancer Research* 84, 2820–2835 (2024). DOI
- Dwivedi P, **Si BR**, Pillai DS, Mangal R. "Solute Induced Jittery Motion of Self-Propelled Droplets." *Physics of Fluids* 33 (2021). DOI

HONOURS AND AWARDS

- **CDI Travel Award**, Dept. of Chemical and Biological Engineering, Johns Hopkins University (2023)
- **Doctoral Fellowship**, Dept. of Chemical and Biological Engineering, Johns Hopkins University (2021-2022)

CONFERENCE PRESENTATIONS

4 presentations at AIChE and BMES Annual Meetings, 2022–2025 (2 oral, 2 poster).

- AIChE Annual Meeting 2025, oral (341c). "Ion Channel-Mediated Regulation of Confined Cell Migration Under Elevated Extracellular Viscosity." Program
- BMES Annual Meeting 2025, poster (K16). "Extracellular fluid viscosity regulates distinct ion channel dynamics to enhance 3D cancer cell migration." Program
- BMES Annual Meeting 2024, poster (S8). "A Microfluidic Model for Electrotaxis in Tumor Microenvironments: Phenotypic Modulations and Calcium Signaling." Program
- AIChE Annual Meeting 2023, oral (604c). "A Novel Microfluidic Device to Study the Influence of Electric Field on Cancer Cell Motility Under Confinement and Its Underlying Effect on Cellular Contractility." Program

TEACHING AND MENTORING

- Teaching Assistant, Advanced Transport Phenomena, Johns Hopkins (2 semesters); Polymer Physics and Chemical Engineering Communication, IIT Kanpur.
- Mentored junior researchers in experimental design, microscopy, and analysis, and coordinated cross-disciplinary collaborations at Johns Hopkins; Undergraduate Academic Mentor, IIT Kanpur.

RESEARCH INTERESTS

Expansion microscopy, microfluidics, hydrogel and polymer-network chemistry, quantitative fluorescence microscopy, optogenetics, tissue and cell staining, image-based quality control, laboratory automation.